

[ask] [en] Last updated: 2026-09-01

What it does: Compares base, low-temperature, and alpha=4 power sampling on MATH500 with frozen Qwen2.5-Math-1.5B; reports accuracy, paired uncertainty, cost, and changed answers.

What it saves: Executed tables and figures inside `notebooks/results.ipynb` ; standalone figures in `../figures/` are produced by `../analyze.py` .

What it uses: `../results/*.json` , `../analyze.py` , and `../math500_sampling/paper_grading.py` ; no model or GPU is loaded.

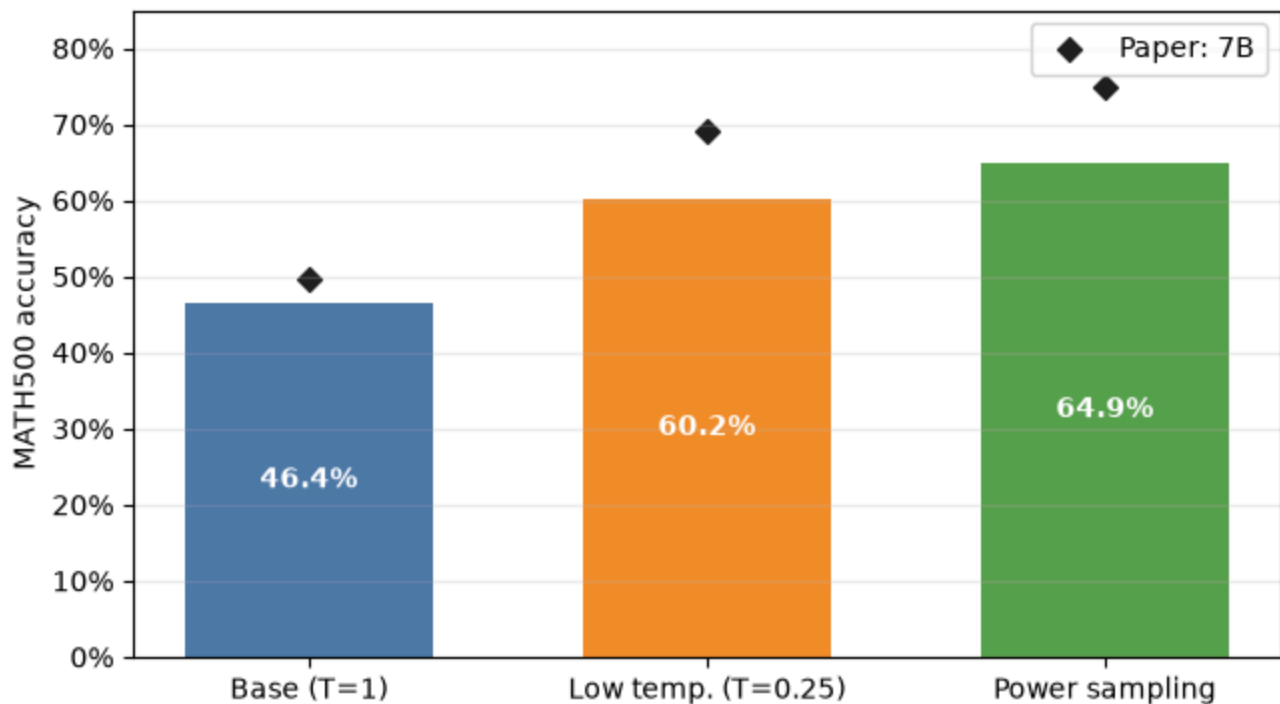
Power sampling on MATH500

The paper samples from the sequence-level distribution $p(\text{response})^\alpha$ using suffix-resampling Metropolis–Hastings. We keep its MATH500 prompt and settings—alpha 4, proposal temperature 0.25, 16 blocks of 192 tokens, 10 updates per block, and 3,072 maximum tokens—but use Qwen2.5-Math-1.5B instead of 7B.

The primary metric uses a local reimplementaion of the paper's grader. `math_verify` is retained as a sensitivity check.

Main result

	seeds	accuracy	math_verify_accuracy	parse_rate
base	[0, 1]	46.4%	47.0%	84.9%
low_temp	[0, 1]	60.2%	61.4%	89.1%
power	[0, 1]	64.9%	66.1%	88.9%



	seeds	effect_pp	95% CI (pp)
low_temp_minus_base	[0, 1]	+13.8	[10.1, 17.4]
power_minus_base	[0, 1]	+18.5	[14.7, 22.3]
power_minus_low_temp	[0, 1]	+4.7	[1.8, 7.7]

Power sampling improves over low-temperature sampling on the observed seed(s). The paired bootstrap resamples questions; with few decoding seeds it does **not** quantify seed-to-seed uncertainty.

Sampling cost and chain behavior

	response_tokens	sampled_tokens	acceptance_rate	sampling_overhead
base	664	664	0.0%	1.0x
low_temp	653	653	0.0%	1.0x
power	628	9861	59.7%	15.7x

Power sampling is not a matched-compute baseline: it returns one answer but samples many suffix tokens. The relevant comparison is single-shot answer quality, with sampled-token overhead reported explicitly.

What changed on the same questions?

	Power	False	True
Low temp.			
False		130	67
True		44	259

	id	target	low-temp answer	power answer	low correct	power correct
0	468	8	7	8	False	True
1	58	9901	10,000	9901	False	True
2	457	0	NaN	0	False	True
3	433	.35625	0.35625	NaN	True	False
4	262	4	4	NaN	True	False
5	12	284	284	NaN	True	False

The paired examples show two mechanisms. Power sampling sometimes replaces a flawed derivation (an off-by-one count or false algebraic identity) with a correct one. It also changes termination and format compliance: some gains come from adding the required box, while some losses are mathematically correct but omit it. See `../QUALITATIVE_EXAMPLES.md` for the full case discussion.

Conclusion and limits

The two-seed 1.5B study reproduces the paper's ordering—power sampling > low temperature > base—without training or changing model weights. This supports the narrower claim that sequence-level sharpening can improve this model's sampled MATH500 answers. It does not establish equivalence to a larger model or RL, and the large generation overhead makes it a quality–compute trade-off rather than a free improvement.